Model page: explicit interfaces

This synthetic page explains what is passed between questions before presenting an algorithm. Figure 1 distinguishes a physical state from a policy decision; the hourly grid and units remain unchanged at the interface.

$$\text{stock}(t) = 100\exp(-kt), \qquad 0.15 \le k \le 0.25. \tag{1}$$

The rate has units of inverse hours. The parameter range expresses a scenario assumption, without assigning probabilities. Equation (1) supplies the state, not the optimization objective.

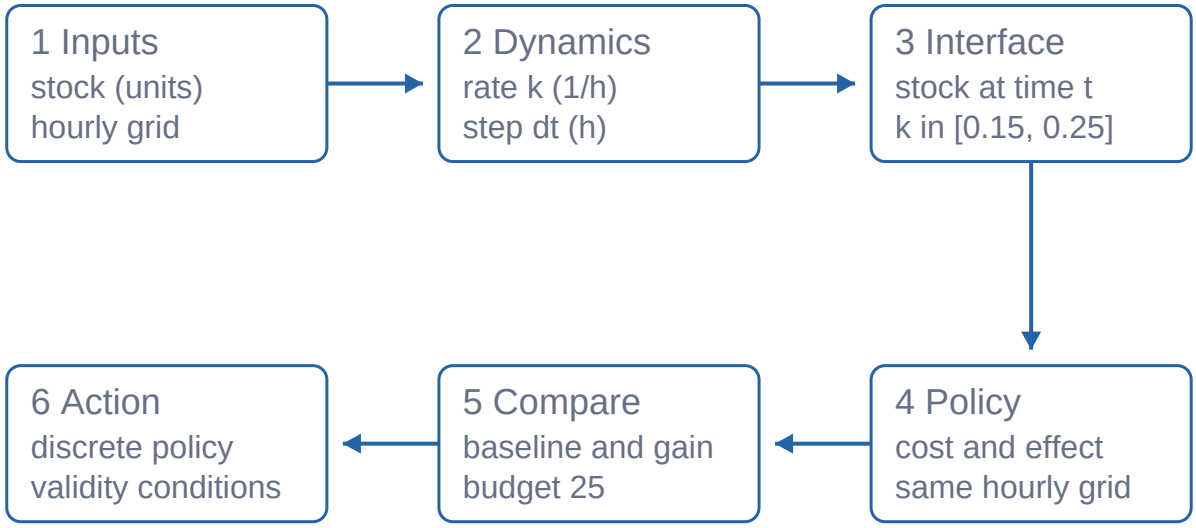


Figure 1: Original synthetic workflow. The mechanism supplies stock on the same hourly grid used by the policy model. The interface carries units and a parameter range; comparison precedes the budget-constrained action. Arrows represent data dependencies, not a claim of numerical validation.

Quantity	Unit	Interface meaning
Stock	units	State at the shared hour
Rate	inverse hours	Bounded model parameter
Cost	cost units	Policy objective

Result page: comparison and magnitude

Both policies use the same nine loads. The mean delay saved is 3.40 minutes across these deterministic scenarios. Figure 2 separates the performance comparison from the magnitude of the paired gain; it does not establish performance on an external population.

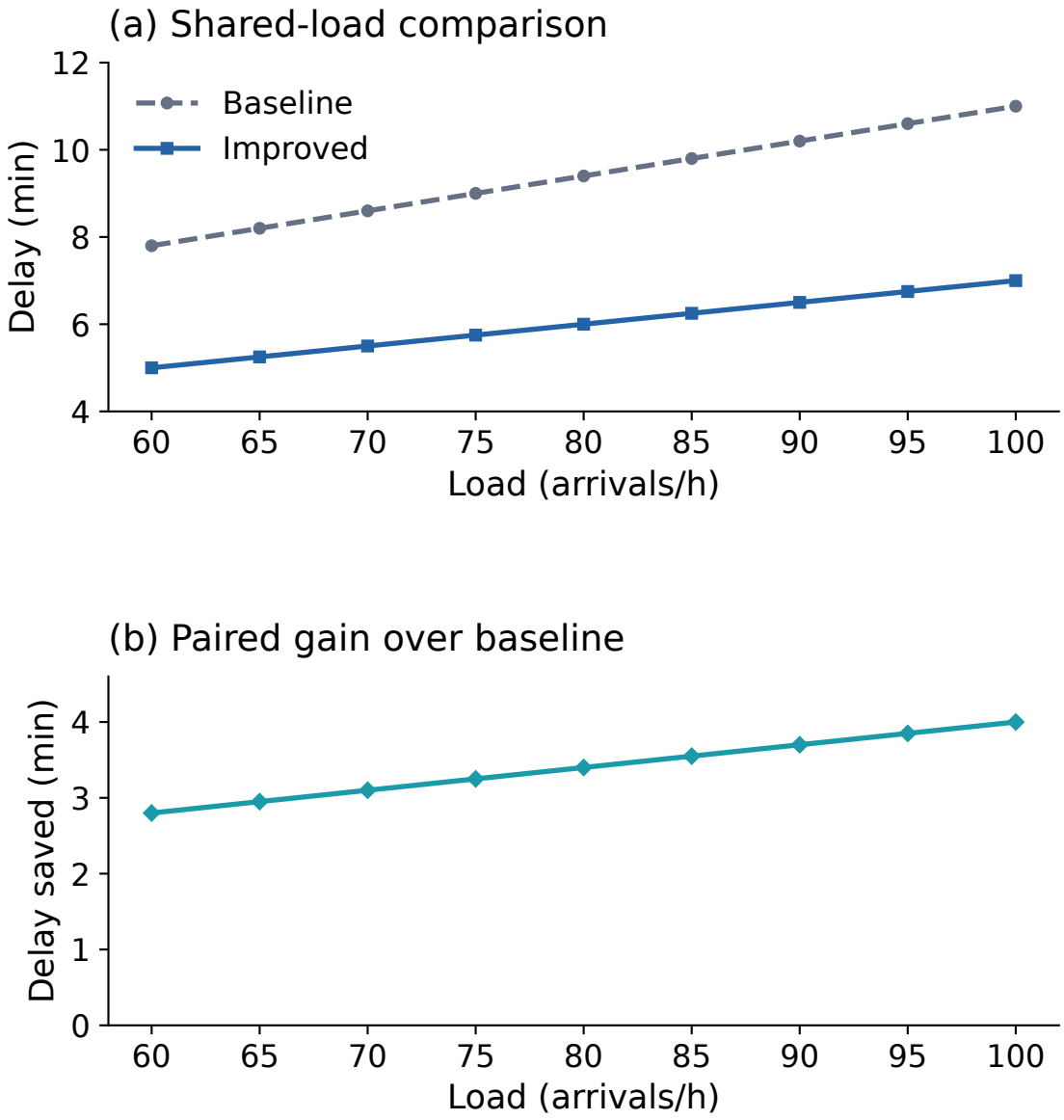


Figure 2: Synthetic paired comparison at nine identical hourly loads. Panel a shows delay for both policies; panel b shows baseline minus improved delay at each load. Every generated scenario is retained. The curves are deterministic illustrative formulas, not estimates from observed service data.

Result page: show the failure window

A fixed synthetic prediction follows the reference until an imposed shift at hour 18. Figure 3 makes the missed regime visible in both the trajectories and signed error. The failure exceeds the stated five-unit tolerance, so an unconditional reliability claim would be unsupported.

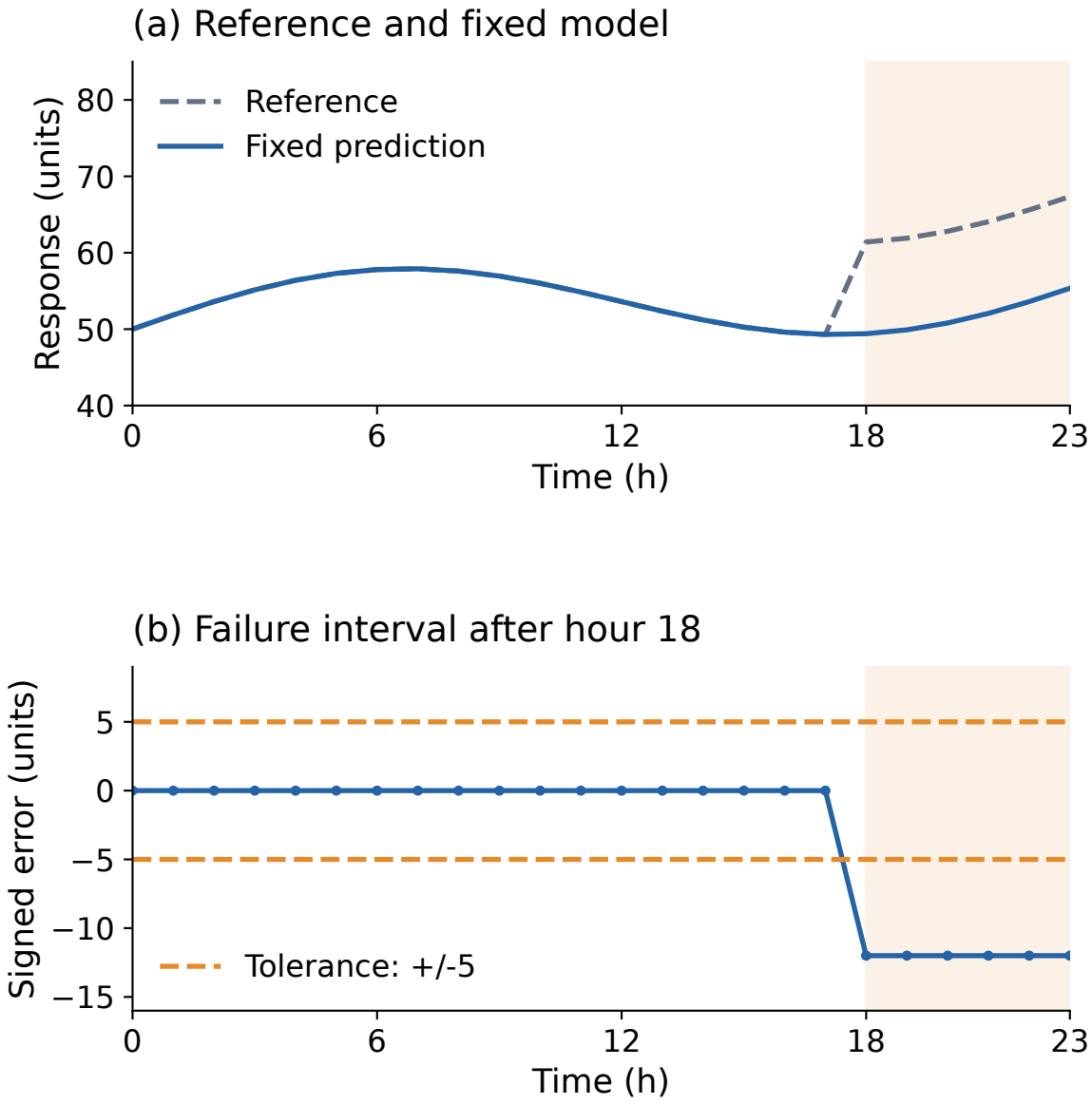


Figure 3: Synthetic reference and fixed prediction over 24 hours. A twelve-unit shift is deliberately introduced at hour 18. Panel a shows trajectories; panel b shows prediction minus reference, with a five-unit tolerance. The shaded interval is a known failure scenario, not a fitted confidence interval or evidence of real-world generalization.

Sensitivity page: numerical and action changes

Two cost functions cross at $p = 14/9$, approximately 1.556. Figure 4 connects the continuous responses to a discrete action change. A small cost variation near the crossing can change the recommendation even when both response curves remain smooth.

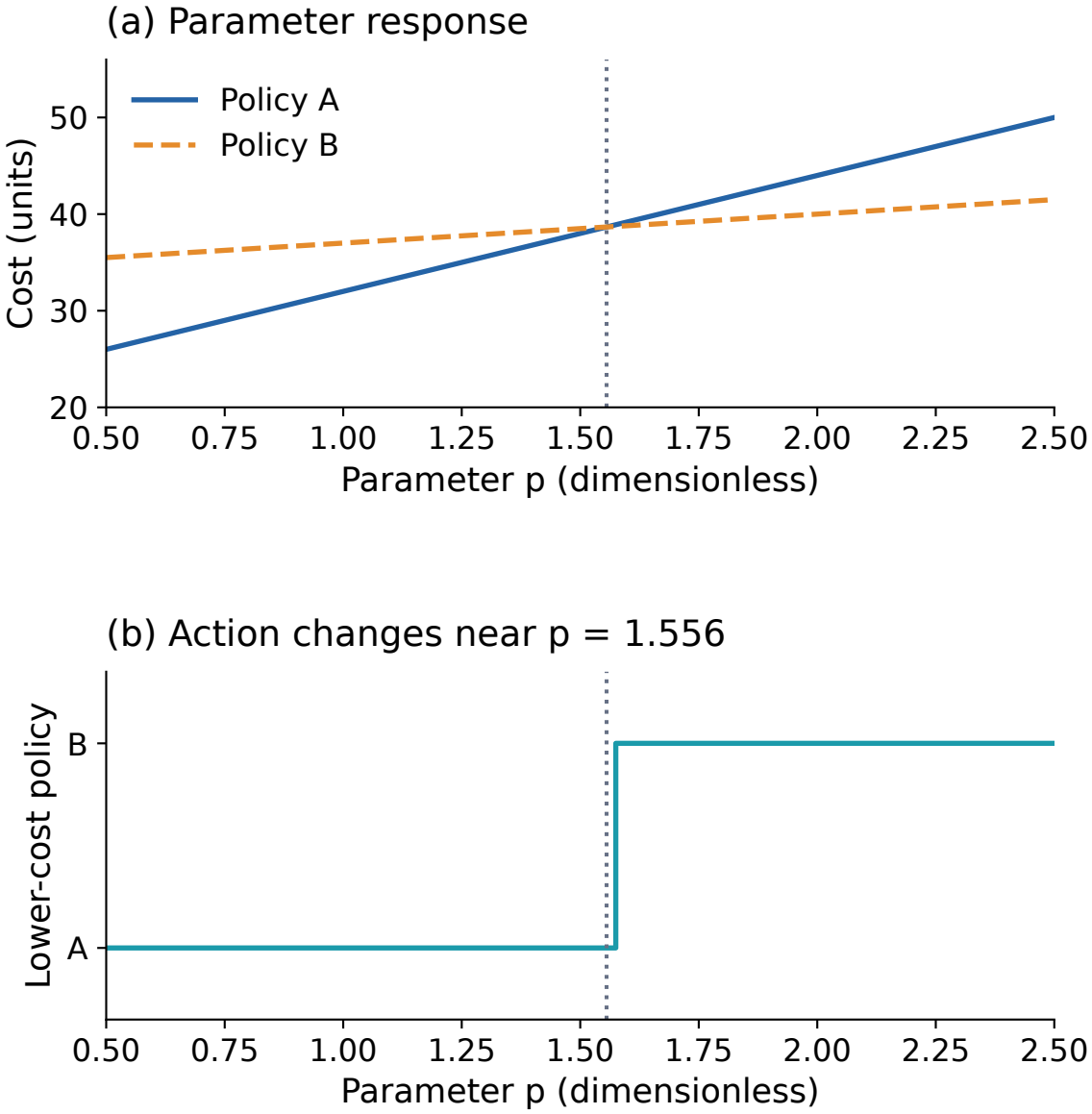


Figure 4: Synthetic cost sensitivity for two policies across 81 parameter values. Panel a shows both cost functions; panel b maps each value to its lower-cost policy. The analytical crossing is 14/9. The example distinguishes a smooth numerical response from a discrete change in the recommended action.

Sensitivity page: identify the uncertainty object

Figure 5 gives two deliberately different constructions. A range over assumed rates is not a confidence interval. The central predictive interval has a probability interpretation only conditional on the stated Normal response model; no empirical coverage claim is made.

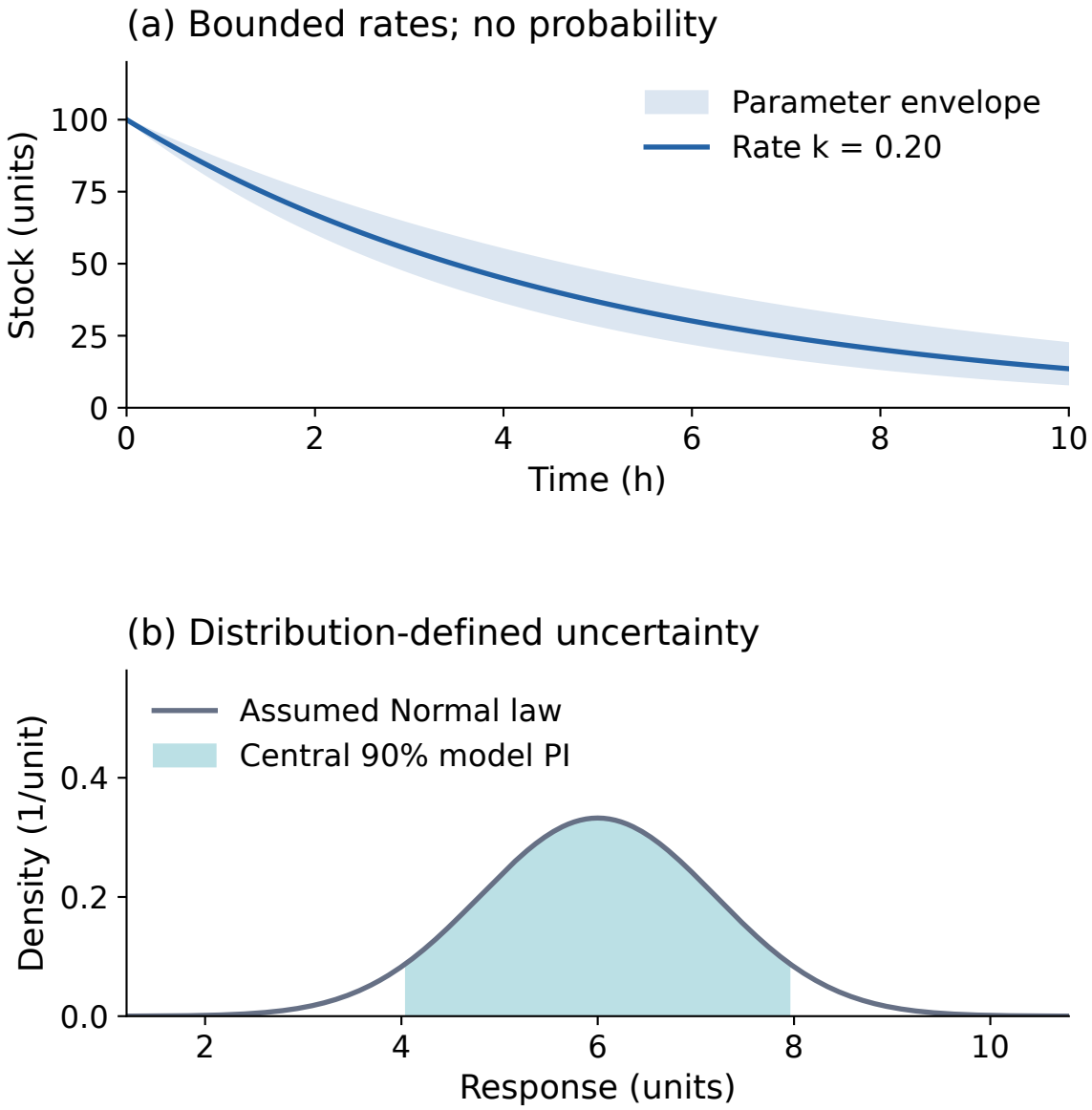


Figure 5: Two synthetic uncertainty constructions. Panel a spans decay rates from 0.15 to 0.25 per hour, without a probability interpretation. Panel b shades the central 90% interval of an assumed Normal response with mean 6 and standard deviation 1.2. This model-defined interval is not empirical coverage validation or a confidence interval for the mean.

Decision page: a conditional recommendation

Among the seven discrete policies, D maximizes effect subject to budget 25. Its cost is 23 and effect is 70. Figure 6 displays both the trade-off and the feasibility reason. The choice is conditional on this candidate set, objective and budget.

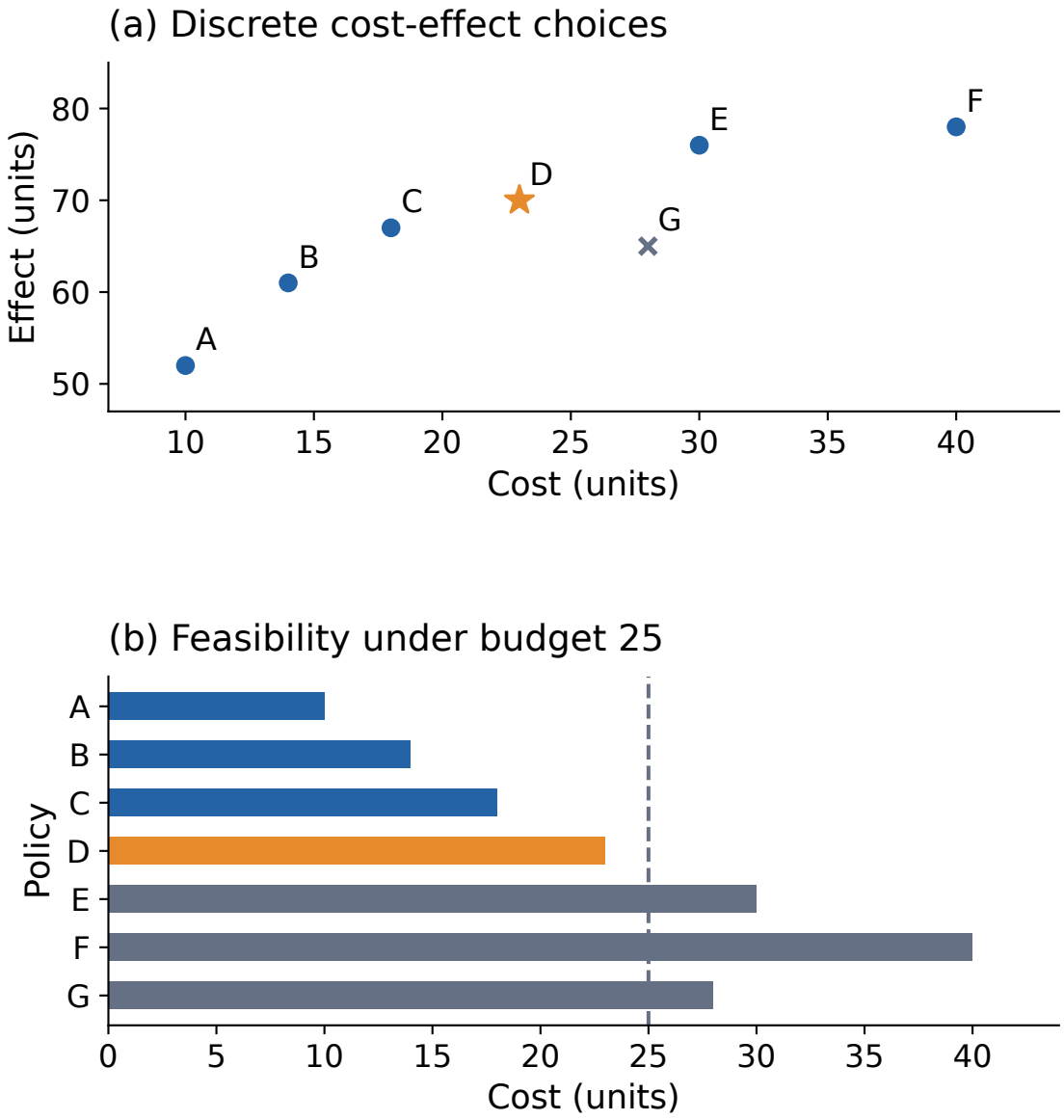


Figure 6: Seven synthetic discrete policies. Panel a shows cost and effect: circles are frontier policies, the cross is dominated G, and the orange star is D. No interpolated actions are implied. Panel b shows costs against budget 25. D is preferred only for this budget and objective.